Phonological Transparency in the Sonority Hierarchy

Revisiting Cross-Linguistic Variations of /v/ in Greek, Russian, and Serbian Using a Sonority-Based Measure Cepstral Peak Prominence (CPP)

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1 Introduction

1.1 Phonological typology of /v/

Voiced fricatives –involving two conflicting articulatory mechanisms– present an inherent production challenge that gives rise to diverse patterns of phonetic variation. The periodic voicing component requires a pressure gradient across the glottis, whereas the aperiodic frication component demands the same force at the locus of supra-laryngeal constriction (Bjorndahl, 2022). Such an aerodynamic tension often leads to the weakening of either vocal fold vibration (voicing) or turbulent airflow (frication). This trade-off is particularly unstable for **spirant /v/**, which has less salient fricative energy than sibilant /z/, whose strong frication more readily leads to devoicing.

The phonetic variation of /v/ is further reflected in its phonological patterning. On the one hand, when /v/ undergoes **devoicing**, it maintains fricative-like properties and thus patterns as an obstruent, distributing as a counterpart to /f/. Greek /v/ constitutes such a case, as evidenced by several distributional facts. First, both /f/ and /v/ can occur as the first member of a homorganic consonant cluster, forming minimally contrastive pairs. Second, within-word voicing agreement shows that /f/ and /v/ alternate in accordance with voicing assimilation (e.g., [evɣloˈtos] 'eloquent' vs. [efstaˈhia] 'steadiness') (Bjorndahl, 2022).

On the other hand, /v/ may undergo **defrication**, maintaining voicing while patterning as an approximant. Serbian /v/ exemplifies this behavior, as evidenced by its distribution. In contrast to the Greek case, Serbian /v/ occurs as the second member of a homorganic consonant cluster. Furthermore, it does not trigger regressive voicing assimilation, suggesting that it does not pattern

with obstruents. For instance, /s̲-gaziti/ → [z̲gaziti] 'trample', but /svariti/ → [svariti] 'digest' (Bjorndahl, 2022).

Meanwhile, as noted, the voicing–frication trade-off in /v/ is unstable, and there are cases in which its phonology embodies **hybrid traits** of both obstruents and sonorants. Russian /v/ constitutes such a case. On the one hand, its participation in final devoicing and regressive voicing assimilation—properties also exhibited by obstruents such as /z/ in Serbian—suggests an obstruent-like status. On the other hand, its failure to trigger voicing assimilation, points toward a sonorant-like behavior (Bjorndahl, 2022).

Thus, Greek, Russian, and Serbian represent a three-way typological division in the phonological behavior of /v/, ranging from more obstruent-like to more sonorant-like patterns (Fig. 1).

1.2 Transparency Hypothesis

A research question raised in Bjorndahl (2022), originally formulated by Padgett (2002), concerns whether the typological divide observed in phonology is *transparently* reflected in phonetic variation. This appears to be a reasonable conjecture, given that phonological variation originate from the aerodynamic tension phonetically inherent in the voiced spirant /v/.

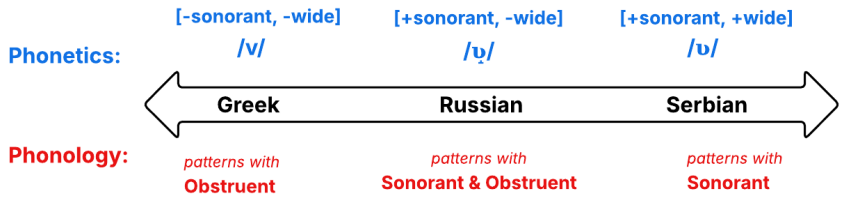

Figure 1: Typological continuum of /v/ at the Phonetics-Phonology interface

If obstruent- and sonorant-like behavior precisely coincide with the phonetic realization of devoicing and defrication, Greek /v/ would bear the feature specifications [-sonorant, -wide], whereas Serbian /v/ would bear [+sonorant, +wide]. The feature [wide], proposed in Padgett (2002), encodes degree of constriction and serves to capture the intermediate status of Russian /v/. Since more fricated, obstruent-like segments exhibit a higher degree of constriction to generate greater turbulence, Greek /v/ can be characterized as [-wide], while Serbian /ʋ/ can be characterized as [+wide]. Russian /ʋ˗/[1] occupies an intermediate position: while it is [+sonorant], it is not fully [+wide], reflecting its

[1]The superscript minus in [ʋ˗] indicates a narrower constriction, used as a notational substitute due to limitations

status as a narrow approximant.

2 Methods

2.1 Previous Approaches

To test the hypothesis outlined in Fig. 1, Bjorndahl (2022) employs four acoustic measures—duration, center of gravity (COG), spectral energy differences (SED), and harmonic-to-noise ratio (HNR)—across recordings of /v/ from three languages under two syllabic conditions: word-initial stressed (WIS) and word-medial unstressed (WMU). The syllabic environment was included as an independent variable, as voiced spirants have been reported to vary in the degree of defrication, with greater reduction in prosodically weak positions such as WMU. That is, they tend to surface as more approximant-like (i.e., more lenited) in intervocalic contexts.

Category formation was assessed through statistical comparisons across language pairs. Specifically, if /v/ in languages A (e.g., Greek) and B (e.g., Serbian) belong to distinct phonetic categories, their values along the acoustic dimensions described above should exhibit statistically significant differences. Under a fully transparent phonology–phonetics mapping, we would expect the pattern shown in Table 1, where checkmarks denote statistically significant differences.

	WIS				WMU			
	Dur.	Har.	CoG	SED	Dur.	Har.	CoG	SED
Greek vs. Serbian	✓	✓	✓	✓	✓	✓	✓	✓
Greek vs. Russian	✓	✓	✓	✓	✓	✓	✓	✓
Russian vs. Serbian	✓	✓	✓	✓	✓	✓	✓	✓

Table 1: Cross-linguistic comparison under a fully transparent phonology–phonetics mapping.

However, such an idealized three-way divide is difficult to obtain because traditional acoustic measures tend to capture either aperiodic spectral properties or periodic voicing properties—an approach that does not suffice to account for the dual aspects of the aerodynamic tension underlying /v/. Because voiced fricatives involve energy distributed across both lower frequencies, associated with sonorancy, and higher frequencies, associated with turbulent airflow, an appropriate measure for investigating their cross-linguistic variation should capture both aspects comprehensively.

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Among the measures employed in Table 1, COG and SED index aperiodic properties, while duration reflects periodic properties. To briefly summarize the expected patterns, COG and SED values are predicted to be lower for more approximant-like realizations of /v/, as increased voicing energy in lower frequencies shifts the concentration of spectral energy downward and reduces high-frequency energy. Duration is expected to be greater for more fricative-like realizations accompanying devoicing, as voiceless frication produces sustained turbulent noise. Thus, all three measures should ideally yield the hierarchy Greek > Russian > Serbian.

	WIS				WMU			
	Dur.	Har.	CoG	SED	Dur.	Har.	CoG	SED
Greek vs. Serbian	✓		✓	✓	✓		✓	✓
Greek vs. Russian					✓		(✓)	✓
Russian vs. Serbian			✓	✓				

Table 2: Cross-linguistic comparison of /v/ reported in (Bjorndahl, 2022)

This expectation is only partially borne out in the results reported by (Bjorndahl, 2022), reproduced in Table 2. Greek, which occupies the most obstruent end of the continuum, shows the highest values across COG, SED, and duration, with statistically significant differences relative to the other two languages. Serbian, at the most sonorant end of the spectrum, contrasts with Greek across all three measures, yielding a robust Greek–Serbian distinction. In contrast, the behavior of Russian varies by prosodic context, suggesting a pattern of **context-dependent categorization**. In the word-initial stressed (WIS) condition, Russian patterns with Greek, as indicated by the lack of significant differences in the Greek–Russian comparison. In the word-medial unstressed (WMU) condition, however, Russian patterns more closely with Serbian.

This raises the question of whether such context-dependent categorization reflects the true phonetic behavior of /v/, or whether it arises from the limitations of acoustic measures that capture only one aspect of the aperiodic–periodic distinction. Admittedly, this contextual pattern is phonetically plausible, as it aligns with the tendency toward increased lenition in prosodically weak intervocalic positions.

In principle, HNR, which measures the ratio between harmonicity—associated with voicing—and noise—associated with frication (see Equation 1)—should provide insight into this question. Among the four measures employed in Bjorndahl (2022), HNR is the only one that conceptually functions as a hybrid metric, capturing both periodic and aperiodic aspects of the voicing–frication dynamics that

characterize /v/. However, this measure shows no discriminative power across the three languages, as evidenced by the absence of checkmarks in the HNR column in Table 2. The author attributes this limitation to the formulation of HNR in Equation 1, in which the noise component in the denominator is defined as the complement of the harmonic component in the numerator. As a result, this *interdependence* obscures the metric's ability to discriminate across languages, particularly when one component is uniformly prevalent. In the present case, Greek, Serbian, and Russian are all true voicing languages, exhibiting similar underlying voicing specifications (Bjorndahl, 2022).

$$\text{HNR (in dB)} = 10\log_{10}\left(\frac{\underbrace{r'_x(\tau_{\max})}_{\text{harmonicity}}}{\underbrace{1 - r'_x(\tau_{\max})}_{\text{noise}}}\right) \tag{1}$$

2.2 Proposal

The limitations of previous work and the questions addressed in this study can be framed at two levels. First, the aeroacoustic dynamics of voiced fricatives require a comprehensive measure that captures both spectral and voicing properties, whereas most traditional acoustic metrics target only one of these dimensions. Second, although HNR conceptually integrates both aspects, its formulation imposes interdependence between harmonic and noise components, limiting its discriminative power across languages. By contrast, measures that target a single dimension (e.g., COG, SED, and duration) have yielded apparent categorization patterns, but these are contingent on prosodic context rather than reflecting stable, context-independent distinctions.

Against this backdrop, the present study seeks to identify a metric that satisfies these desiderata and to evaluate whether cross-linguistic categorization can be achieved independently of prosodic context, or whether it remains conditioned by it.

To identify an appropriate metric, the present study draws on the analysis of sonority proposed in Miller (2012). The typological continuum in Fig. 1 can be recast along a sonority scale, with the Greek [-sonorant, -wide] endpoint corresponding to lower sonority values and the Serbian [+sonorant, +wide] endpoint to higher values. A key insight of Miller (2012) is that sonority can be understood as a composite measure consisting of two scalar components: the source and the aperture scales. These correspond to the periodic and aperiodic aspects of the acoustic signal, respectively, which are precisely the dimensions targeted in the present study.

Paralleling the source–filter model, the source scale reflects the nature of the sound source, while the aperture scale is inversely related to the degree of attenuation resulting from impedance in the supralaryngeal vocal tract. The hierarchies defined in Equation 2 jointly determine the values of sonority, while functionally *independent* from one another. Scalar values are indicated alongside each element in the respective hierarchies. Under this hierarchy, devoiced realizations of /v/ correspond to values of (2, 3), whereas defricated realizations correspond to values of (4, 5), placing the latter at the higher end of the spectrum in Fig. 1.

$$\begin{aligned} &\text{Source scale:} \quad \text{NS (no source) } \mathbf{1} < \text{TS (turbulence only) } \mathbf{2} < \text{PS+GS (breathy voice) } \mathbf{3} < \text{PS (modal voice) } \mathbf{4} \\ &\text{Aperture scale:} \quad \text{stops } \mathbf{1} < \text{implosives } \mathbf{2} < \text{fricatives } \mathbf{3} < \text{nasals } \mathbf{4} < \text{liquids/glides } \mathbf{5} < \text{vowels } \mathbf{6} \end{aligned} \tag{2}$$

2.3 Implementation

While the study's desideratum of a comprehensive, non-interdependent measure aligns with the analysis of sonority in Miller (2012), Equation 2 remains a formal characterization. An acoustic metric that operationalizes this insight is therefore required. To this end, we propose cepstral peak prominence (CPP), a measure widely used in clinical phonetics (Fraile and Godino-Llorente, 2014).

CPP operates in the quefrency domain, where source-related periodicity and filter-related spectral structure are jointly represented yet remain separable. This property directly addresses the need for a measure that captures both dimensions without introducing interdependence. Formally, the cepstrum is obtained by taking the Fourier transform $\mathcal{F}$ of the logarithm of the magnitude spectrum $\log |S(f)|$. The logarithmic transformation converts the convolutional interaction between the glottal source and the vocal tract filter into an additive relationship. As shown in Fig. 2, this transformation enables the two components to be disentangled.

In the cepstral domain, periodic source excitation is mapped to distinct peaks at higher quefrencies, whereas the spectral envelope appears as a slowly varying component at lower quefrencies. As a result, periodicity and spectral structure are encoded in different regions of the same representation, enabling their simultaneous yet independent characterization. Under this representation, CPP quantifies the prominence of the cepstral peak relative to a regression line fitted to the cepstral envelope. In contrast to Equation 1, which expresses the relationship between harmonicity and noise in a ratio form, CPP measures this relationship additively, as shown in Equation 3.

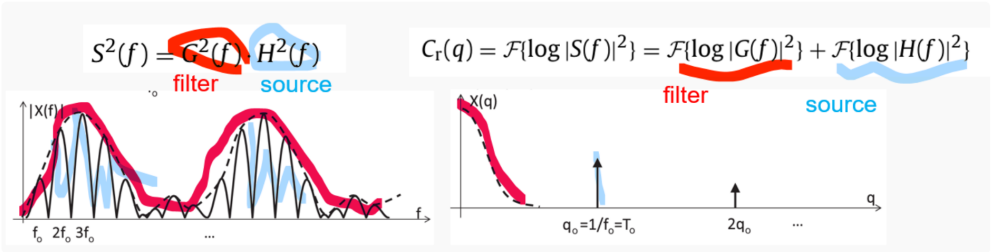

Figure 2: Illustration of cepstrum derivation (adapted from (Fraile and Godino-Llorente, 2014)). S denotes the spectrum, G the glottal signal, H the combined effect of the vocal tract and lip radiation, f frequency, $\mathcal{F}$ the Fourier transform, q quefrency, and C the cepstrum.

$$\text{CPP} = \underbrace{C(q_{\text{peak}})}_{\text{cepstral peak (periodicity)}} - \underbrace{\hat{C}(q_{\text{peak}})}_{\text{regression baseline (aperiodic component)}} \tag{3}$$

Murray et al. (2025) provide a Praat plug-in for calculating CPP, which eliminates the need for complex scripting by offering a user-friendly interface. The program can be downloaded from [2] and installed by placing the folder in the Praat preferences directory; the interface becomes available upon launching Praat. Using this tool, the present study calculates CPP for synthesized speech samples from the three languages investigated in Bjorndahl (2022). Following their configuration, one speech token was generated per language for each of two prosodic conditions—word-initial stressed (WIS) and word-medial unstressed (WMU)—yielding a total of six analyzed sound files. The CPP settings followed standard parameters implemented in the Praat plug-in (e.g., time step = 0.002 s, pitch floor = 60 Hz).

WIS stimuli follow a CV structure, whereas WMU stimuli follow a VCV structure, with vowels restricted to /a/ across all conditions. This restriction follows the experimental design of Bjorndahl (2022), who limited vowel fillers to /a/ and /o/ in order to avoid palatalization effects. The phonetic transcriptions of the analyzed segments, along with their orthographic counterparts used to generate synthetic speech in Google Translate, are provided in Table 3.

Environment	Greek	Russian	Serbian
WIS	/βα/ [va]	/Ba/ [ʊ̞a]	/Ba/ [ʊa]
WMU	/αβα/ [ava]	/aBa/ [aʊ̞a]	/aBa/ [aʊa]

Table 3: Synthetic speech stimuli used for CPP calculation.

[2] https://osf.io/t5hrv/

3 Results and Discussion

Table 4 presents CPP values for the stimuli in Table 3. Given the formulation in Equation 3, more devoiced, fricative-like realizations are expected to yield lower CPP values, whereas more defricated, sonorant-like realizations should yield higher values. The former exhibit a steeper spectral tilt and a less distinct cepstral peak, while the latter exhibit a flatter spectral tilt due to reduced high-frequency energy and a more distinct peak associated with periodicity. These properties respectively raise and lower the regression baseline and lower and increase peak prominence, thereby producing opposite effects. Accordingly, CPP values are predicted to increase along the sonority continuum in Fig. 1, from Greek (least sonorous) to Russian (intermediate) to Serbian (most sonorous), with higher values also expected in the prosodically weak WMU condition than in WIS.

Environment	Greek	Russian	Serbian
WIS	11.71	12.30	13.36
WMU	12.75	10.90	13.85

Table 4: CPP values (in dB) for each language and prosodic condition.

This pattern is largely borne out. Greek and Serbian conform to both cross-linguistic and within-language expectations: CPP increases from Greek to Serbian, and WMU yields higher values than WIS in both languages. Russian follows the same ordering in the WIS condition (Greek $<$ Russian $<$ Serbian), but diverges in WMU, where its CPP value falls below that of Greek. This reversal also disrupts the within-language pattern, as the word-initial stressed position (12.30) exceeds the word-medial position (10.90), contrary to expectation. The resulting asymmetry yields a three-way typological distinction, though its implications for the transparency hypothesis remain uncertain, given that the present analysis is based on synthesized speech and may reflect an outlier in the Russian WMU condition.

Despite this deviation, the broader pattern supports the proposal that CPP functions as a comprehensive measure and a potential substitute for HNR. Its role in establishing phonetics–phonology correspondences, however, warrants further investigation, ideally through corpus-based studies of the three languages considered here and others reported to occupy intermediate positions (e.g., Czech, Bulgarian, Hungarian; Bjorndahl (2022)).

References

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A Spectral and Cepstral Plots

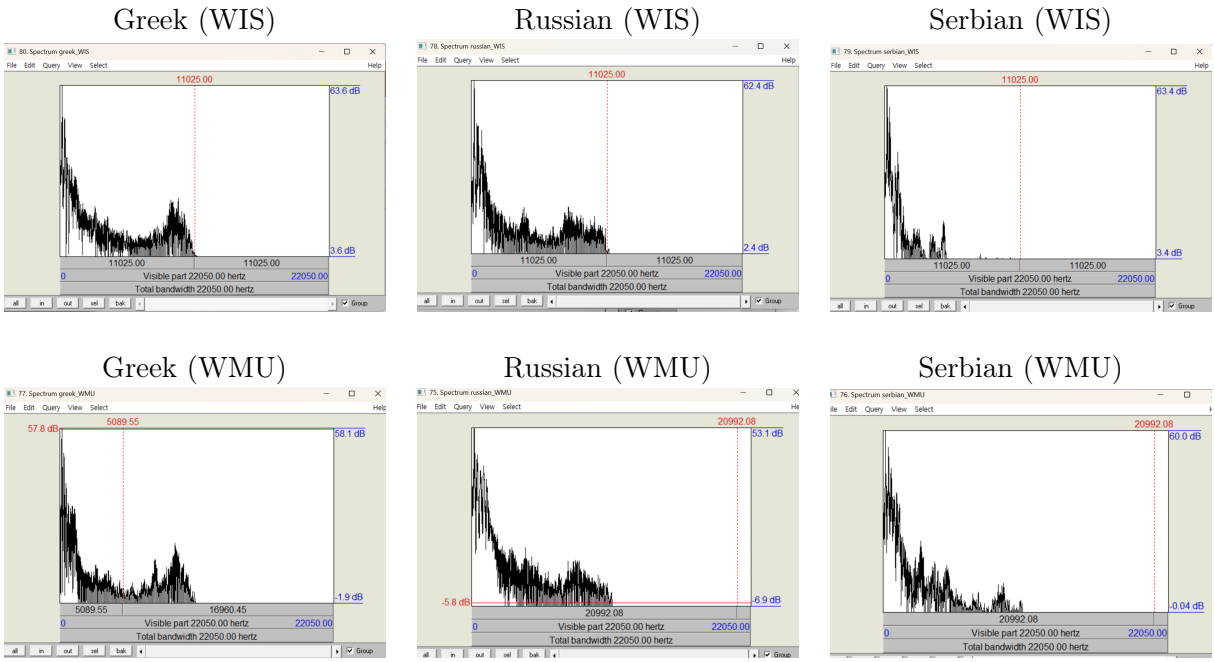

Figure 3: Spectra across languages (columns) and prosodic conditions (rows: WIS vs. WMU)

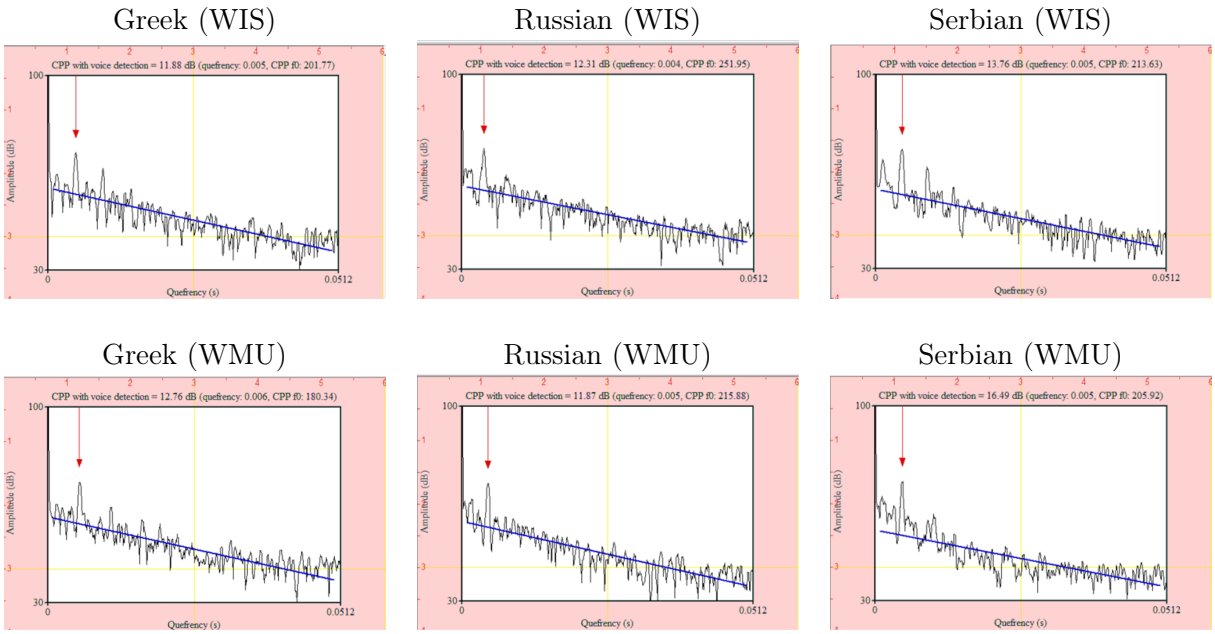

Figure 4: Cepstral representations with CPP measurements across languages (columns) and prosodic conditions (rows: WIS vs. WMU)